

1 Methodology

1.1 Overview

This study proposes an enhanced deep learning framework for cancer survival prediction using multi-omics data, including clinical data, mRNA, miRNA, and CNV. The model improves feature representation, fusion, and prediction by incorporating autoencoders, transformer encoders, contrastive learning, multi-head attention, and a deep survival model (DeepSurv).

1.2 Data Representation Learning

1.2.1 Autoencoder-Based Denoising

Each omics data type is first processed using an autoencoder to remove noise and reduce dimensionality. The encoder learns a compact latent representation:

$$z^{(k)} = \text{Encoder}(x^{(k)}) \quad (1)$$

where $x^{(k)}$ represents the input of the k -th omics data and $z^{(k)}$ is the latent feature vector.

1.2.2 Transformer Encoder

The latent representations are then passed through a Transformer encoder to capture complex relationships:

$$h^{(k)} = \text{Transformer}(z^{(k)}) \quad (2)$$

This helps in learning long-range dependencies and non-linear interactions within each modality.

1.3 Cross-Modal Representation Alignment

1.3.1 Contrastive Loss

To align features from different omics of the same patient while separating different patients, contrastive learning is applied. The loss is defined as:

$$L_{\text{contrastive}} = y \cdot \|h_i - h_j\|^2 + (1 - y) \cdot \max(0, m - \|h_i - h_j\|)^2 \quad (3)$$

where $y = 1$ for the same patient and $y = 0$ for different patients, and m is the margin.

1.4 Multi-Omics Feature Fusion

1.4.1 Multi-Head Attention

To effectively combine multi-omics features, a multi-head attention mechanism is used:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O \quad (5)$$

This allows the model to learn interactions across different modalities and assign importance dynamically.

1.5 Survival Prediction

1.5.1 DeepSurv Model

The fused feature representation is passed to a DeepSurv model, which extends the Cox proportional hazards model using neural networks:

$$r = f(h_{fusion}) \quad (6)$$

The Cox partial likelihood loss is defined as:

$$L_{cox} = - \sum_{i \in E} \left(r_i - \log \sum_{j \in R_i} e^{r_j} \right) \quad (7)$$

where E is the set of observed events and R_i is the risk set.

1.6 Overall Loss Function

The final loss combines survival prediction and representation learning:

$$L = L_{cox} + \lambda L_{contrastive} \quad (8)$$

where λ controls the contribution of contrastive loss.

1.7 Training Strategy

The model is trained end-to-end using backpropagation. All components, including feature extraction, alignment, fusion, and prediction, are optimized jointly using gradient-based optimization (e.g., Adam optimizer).

1.8 Summary of Improvements

The proposed model improves upon traditional approaches by:

- Replacing Siamese loss with contrastive loss for better feature separation
- Using transformer encoders for richer feature extraction
- Applying multi-head attention for advanced feature fusion
- Replacing the linear Cox model with DeepSurv for non-linear survival prediction